EMA705 - Advanced Topics in Finite Elements

HW 4 (Multiphysics)

Given: 4 May., 2026, due: 10 May., 2026. No extensions permitted.

Note: Graded for attempt and submission, so try to complete as much as possible. It's OK if some parts of your implementation are incomplete or incorrect.

Problem

Develop a multiphysics code demonstrating coupled mechanics, thermal diffusion and chemical transport. Assuming a St. Venant-Kirchhoff strain energy density function, and standard Fick's and Fourier's constitutive relations, solve a 2D or 3D initial boundary value problem (IBVP) demonstrating mechanical deformation coupled with temperature evolution and chemical species transport that is relevant to the (highly simplified-) multiphysics observed in Lithium-ion batteries, with the following free energy density expressions and boundary conditions.

- Part 1 : Solve a small-strain or finite-strain mechanics problem with the boundary conditions shown in Figure 1(a). Consider Young's modulus 1.0 N/m^2, and Poisson's ratio 0.3. The free energy density for this part is the classical St. Venant-Kirchhoff strain energy density, and is of the form:
  $\Pi(\boldsymbol{u}) = \frac{1}{2}C_{ijkl}\varepsilon_{ij}\varepsilon_{kl}$

- Part 2 : Now on top of Part-1, add a Fourier heat conduction problem and solve for the coupled evolution of temperature and displacement. The temperature boundary conditions are T=1 $°C$ on the left boundary, and the temperature initial condition is T=0 $°C$ in the interior of the domain, as shown in Figure 1(b). The free energy density for this part is of the form:

  $\Pi(\boldsymbol{u}, T) = \frac{1}{2}C_{ijkl}(\varepsilon_{ij} - \alpha^T_{ij}T)(\varepsilon_{kl} - \alpha^T_{kl}T) + T^2$

  Consider the coefficient of thermal expansion to be α^T_{ij}=0.01 δ_{ij} K^{-1} and thermal conductivity to be D_T=1.0 $W/(m.K)$. Implement a staggered solution scheme with Forward-Euler time stepping.

- Part 3 : Now on top of Part-2, add a Fick's chemical diffusion problem and solve for the coupled evolution of composition, temperature and displacement. The additional composition boundary conditions are c=1.0 on the top boundary, and the composition initial condition is c=0 in the interior of the domain, as shown in Figure 1(c). The free energy density for this part is of the form:

  $\Pi(\boldsymbol{u}, T, c) = \frac{1}{2}C_{ijkl}(\varepsilon_{ij} - \alpha^T_{ij}T - \alpha^c_{ij}c)(\varepsilon_{kl} - \alpha^T_{kl}T - \alpha^c_{kl}c) + T^2 + c^2$

  Consider the coefficient of chemical expansion to be α^c_{ij}=0.005 δ_{ij} and chemical diffusivity to be D_c=1.0 m^2/s. Implement a staggered solution scheme with Forward-Euler time stepping.

Assume zero Neumann boundary conditions on boundaries where no boundary conditions explicitly specified.

Note: Now that you are well versed in FE theory and are getting comfortable with FE coding, the choice of 2D or 3D, time steps, and FE constructs (elements, quadrature, basis order, etc.) is left to your wisdom and convenience. The expectation is that you (1) Clearly code (and identify the sections of the code) that are dealing with the multiphysics equations, (2) Clearly plot the results showing the **time evolution** of displacement, temperature and composition fields.

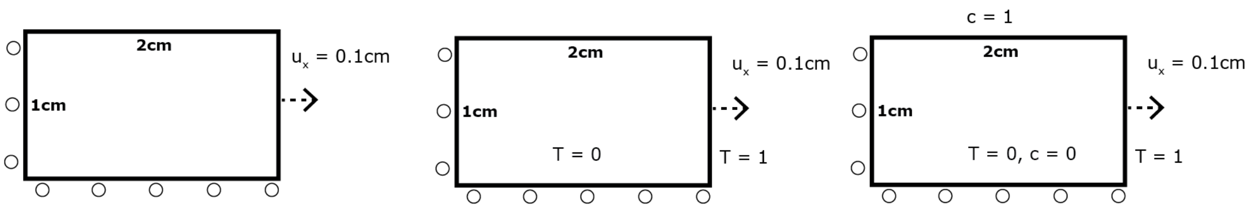

Figure 1 (a) Figure 1 (b) Figure 1 (c)

Submit the source code and a single PDF containing the following plots on Canvas:

- Source code for the 2D FE program. The code should be well commented to clearly show the various steps of the multiphysics residuals, FE solution procedure, time stepping, boundary conditions and initial conditions.
- For Part (1), plot the displacement field over the deformed configuration (deformed mesh).
- For Parts (2) and (3), plot the relevant fields (displacement, temperature and composition) at different points of the time evolution over the deformed configuration showing the deformed geometry and the field contours. Videos of the time evolution of the fields appreciated, but not necessary.